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1) For an n-channel MOS transistor with $\mu_n = 600\ cm^2/V \cdot s$, $C_{ox} = 7 \times 10^{-8}\ F/cm^2$, $W = 20\mu m$, $L = 2\mu m$ and $V_{T0} = 1.0V$, examine the relationship between the drain Current and the terminal Voltage!

Sol: Given Data :

$\mu_n = 600 cm^2/V \cdot s$ $\quad C_{ox} = 7 \times 10^{-8} F/cm^2$

$W = 20 \times 10^{-6} m$ $\quad L = 2 \times 10^{-6} m$

$V_{t0} = 1.0V$

step1: Calculate process Transconductance parameter

$$k_n' = \mu_n C_{ox} = 600 \times 7 \times 10^{-8}$$

$$k_n' = 4.2 \times 10^{-5} A/V^2 \rightarrow (1)$$

step2: Calculate Device TransConductance parameter

$$k_n = k_n' \frac{W}{L} = 4.2 \times 10^{-5} \times 10$$

$$k_n = 4.2 \times 10^{-4} A/V^2 \rightarrow (2)$$

step3: Drain Current Voltage & terminal Voltage

1) Cut-off Region:

$$V_{GS} < V_T \quad [\because I_D = 0]$$

2) Triode linear region

$$V_{GS} > V_T \Rightarrow V_{DS} < V_{GS} - V_T$$

$$I_D = k_n \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$= 4.2 \times 10^{-4} \left[(V_{GS} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right] A$$

c) Saturation Region :

$$V_{GS} > V_T \rightarrow V_{DS} \geq V_{GS} - V_T$$

$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2$$

$$= \frac{1}{2} \times 4.2 \times 10^{-4} (V_{GS} - 1)^2$$

$$I_D = 2.1 \times 10^{-4} (V_{GS} - 1)^2 \ A$$